



Section: Our Dynamic Universe



Topic: The Expanding Universe

Question 6 – Commenting on Song Lyrics

The lyrics in question:

“A fraction of a second and the elements were made.
It’s expanding ever outward but one day
It will cause the stars to go the other way...”

Physics-Based Commentary

✓ 3 marks



Marking-Instruction-Aligned Response:

- The first line refers to the **Big Bang**, where in the first few seconds, **protons and neutrons formed**, followed by **nuclear fusion** creating elements like **helium and lithium**.
- The second line refers to the **observed expansion** of the universe, supported by **redshift** and **Hubble’s Law**.
- The final line suggests a **possible future contraction** (the “Big Crunch”), which is **not currently supported** by observations. Recent evidence (e.g. from **supernova data**) suggests that the universe’s expansion is actually **accelerating**, due to **dark energy**.



Summary:

The lyrics are broadly consistent with current cosmological theory, though the idea of a future contraction is **not consistent** with current observations.



Revision Tips

- The **Big Bang** is supported by:
 - **Cosmic microwave background radiation**
 - **Redshift of galaxies**
- **Nucleosynthesis** occurred within the first few minutes
- **Dark energy** explains the **accelerating expansion** of the universe
- Theories like the **Big Crunch** are speculative and not currently favoured by data